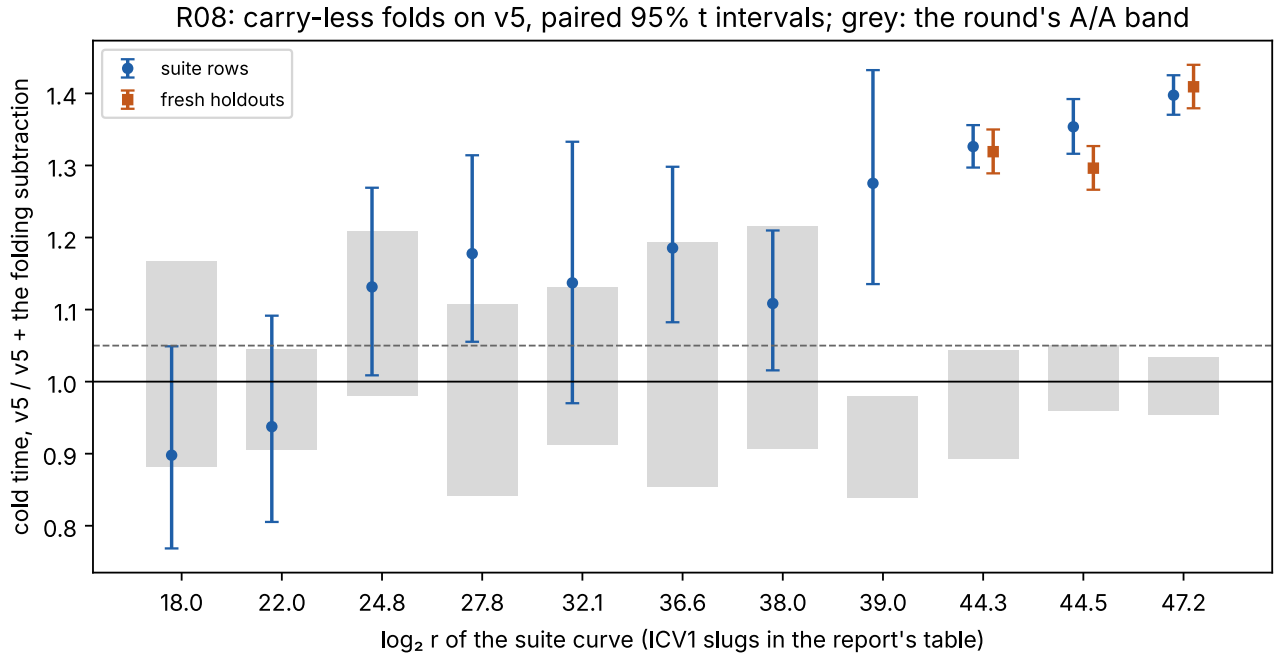
R08: the scan's subtraction by carry-less folds

Accepted, as baseline v6 (engineering). v5 (v4 plus R06′s scan key) against v5 plus R08′s candidate (`candidate-v5.patch`: each product reduced by two `vpclmulqdq` folds, eight lanes; Montgomery's trick in two chains; the negated base as coordinate slices; the build's rows past their orbit on the same kernel), both built with `rustc 1.97.0`, on an Emerald Rapids host of the reference class, 2026-10-16. Protocol: `research/ic_tool_program/rounds/R08-scan-fold/PROTOCOL.md` (declaration, amendment 1); analysis: `analysis.json` (`icprog analyse r08`); run trees: `runs.tar.xz` (SHA-256 `90c32b66b7ae9708…`).

- **Tests:** 206 of 206 `koblitz_fast` and `koblitz_index_calculus` tests on the candidate's tree, among them the folding kernel against scalar addition, the scans' witnesses against the point list's, and the two-pass build table.
- **Pin:** the candidate equals v0′s outputs on all 90 suite rows.
- **Timed:** 1,101 isolated processes (A/A 220, suite 481, holdouts 400), 1 contended and re-run, none failed; the driver task was relaunched once on the same container and resumed from disk.
- **Decision:** at each target size the cold interval lies above 1.05 on the suite rows and on the fresh holdouts; no size regresses beyond its A/A band. Predicted 1.25–1.33, 1.16–1.29 and 1.27–1.46.



research/ic_tool_program/rounds/R08-scan-fold/analysis.json; 1,101 isolated processes, 1 contended (excluded); dashed: the 1.05 acceptance line

Figure 1: Cold time, v5 over the candidate (above 1, the candidate is faster), paired 95% t intervals; grey: the round's own A/A band; dashed: 1.05. Values from `analysis.json`.

Suite rows (five rounds; ten at 2^44.3)

curve	$\log_2 r$	rows	cold [95%]	collection	S cold, v5 → v6	A/A band
`icv1-f2m19-tm797-9c54981b`	18.002	2	0.898 [0.769, 1.049]	0.957	129.11 → 141.28	0.883–1.167
`icv1-f2m23-tm5197-1f85e9e1`	22.001	2	0.938 [0.805, 1.092]	1.011	33.64 → 35.74	0.906–1.045
`icv1-f2m45-tm6236725-40939294`	24.803	2	1.131 [1.009, 1.269]	1.224	24 → 20.75	0.98–1.209
`icv1-f2m37-tm534059-32aad96b`	27.781	2	1.178 [1.055, 1.314]	1.228	11.31 → 9.12	0.841–1.107
`icv1-f2m43-tm998717-e2e742b0`	32.113	2	1.137 [0.97, 1.333]	1.222	5.42 → 4.58	0.912–1.13
`icv1-f2m47-t22705043-f4e44623`	36.636	2	1.185 [1.082, 1.298]	1.401	3.27 → 2.76	0.855–1.193
`icv1-f2m57-tm747311035-c1f545af`	38.002	2	1.108 [1.016, 1.21]	1.235	3.59 → 3.44	0.906–1.215
`icv1-f2m41-tm2308219-7f48b14a`	39.0	2	1.275 [1.135, 1.432]	1.429	4.66 → 3.67	0.838–0.979
`icv1-f2m53-tm56619371-dac20a85`	44.259	8	1.326 [1.297, 1.356]	1.468	5.48 → 4.12	0.894–1.044
`icv1-f2m59-tm943548413-98844ecc`	44.517	8	1.354 [1.316, 1.392]	1.529	5.19 → 3.9	0.96–1.051
`icv1-f2m61-t158598901-ab42b6c5`	47.211	8	1.398 [1.37, 1.425]	1.486	7.08 → 5.02	0.954–1.034

Fresh holdouts (seeds 222–225, T135–T142; ten rounds at 2^44.3 and 2^44.5)

curve	$\log_2 r$	pairs	cold [95%]	S cold, v5 → v6
`icv1-f2m53-tm56619371-dac20a85`	44.259	80	1.319 [1.289, 1.35]	5.25 → 3.94
`icv1-f2m59-tm943548413-98844ecc`	44.517	80	1.296 [1.266, 1.327]	4.99 → 3.83
`icv1-f2m61-t158598901-ab42b6c5`	47.211	40	1.409 [1.379, 1.44]	6.92 → 4.96

Scope

One host and one core, the reference class; elsewhere, and in any field that does not fold, the candidate runs v5′s code and no claim is made. The field elements are identical by construction and by test, so the counts and the ratio to the floor are unchanged. At the two smallest sizes the point estimates read 0.90 and 0.94, with intervals of ±14–15% that reach above 1 and stay inside the A/A rule. The distance to matched rho is not re-measured here: a rule run at v6 re-measures it.